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Lab:3

Relational Operators, Logical Operators and Conditionals in C++

Objective of the Lab

The objective of this lab is to understand **relational operators, logical operators, and conditional statements in C++**. These concepts allow programs to make **decisions based on conditions**. By using these operators and statements, a program can compare values and execute different instructions depending on the result of the condition.

Introduction

In programming, many situations require a program to make decisions. For example, a program may need to check whether a number is positive or negative, or whether a person is old enough to vote.

C++ provides **relational operators, logical operators, and conditional statements** to handle such situations.

Relational operators are used to **compare two values**, logical operators are used to **combine multiple conditions**, and conditional statements such as **if, if-else, and else-if** are used to control the flow of the program.

Basic Concepts

1. Relational Operators

Relational operators compare two values and return **true or false**.

Operator	Meaning
==	Equal to
!=	Not equal to
>	Greater than
<	Less than
>=	Greater than or equal to
<=	Less than or equal to

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2. Logical Operators

Logical operators are used to combine multiple conditions.

Operator	Meaning
&&	Logical AND
||	Logical OR
!	Logical NOT

Example:
If we want to check whether a number is **between 0 and 100**, we use:

```
number > 0 && number < 100
```

---

3. Conditional Statements

Conditional statements allow a program to make decisions.

Common conditional statements include:

- **if statement**
- **if-else statement**
- **else-if statement**

These statements evaluate conditions and execute code accordingly.

Task 1: Comparison / Relational Operators

Program

```
#include <iostream>
using namespace std;

int main()
{
    int num1, num2;

    cout << "Enter first number: ";
    cin >> num1;

    cout << "Enter second number: ";
    cin >> num2;

    if(num1 == num2)
        cout << "Both numbers are equal.";

    if(num1 > num2)
        cout << "First number is greater than second.";

    if(num1 < num2)
        cout << "First number is less than second.";

    return 0;
}
```

Explanation

This program asks the user to enter two numbers.
The relational operators **==, >, and** < are used to compare the numbers.

Depending on the comparison result, the program displays an appropriate message indicating whether the numbers are **equal, greater, or smaller**.

Task 2: Logical Operators

Program

```
#include <iostream>
using namespace std;

int main()
{
    int number;

    cout << "Enter an integer: ";
    cin >> number;

    if(number > 0 && number < 100)
        cout << "The number is between 0 and 100.";
    else
        cout << "The number does not meet the conditions.";

    return 0;
}
```

Explanation

This program checks whether the entered number is **greater than 0 and less than 100**.

The **logical AND operator (&&)** is used to combine both conditions.
If both conditions are true, the program prints that the number is within the range. Otherwise, it displays a different message.

Task 3: If Statement

Program

```
#include <iostream>
using namespace std;

int main()
{
    int number;

    cout << "Enter an integer: ";
    cin >> number;

    if(number > 0)
        cout << "The number is positive.";

    return 0;
}
```

Explanation

In this program, the **if statement** checks whether the entered number is **greater than zero**.

If the condition is true, the program prints that the number is positive.
If the condition is false, the program simply ends without printing anything.

Task 4: If-Else Statement

Program

```
#include <iostream>
using namespace std;

int main()
{
    int age;

    cout << "Enter your age: ";
    cin >> age;

    if(age >= 18)
        cout << "You are eligible.";
    else
        cout << "You are not eligible.";
```

```
    return 0;
}
```

Explanation

This program asks the user to enter their age.

The **if-else statement** checks whether the age is **greater than or equal to 18**.

- If the condition is true, the program prints that the person is eligible.
- If the condition is false, it prints that the person is not eligible.

Task 5: Else-If Statement

Program

```
#include <iostream>
using namespace std;

int main()
{
    int number;

    cout << "Enter an integer: ";
    cin >> number;

    if(number > 0)
        cout << "The number is positive.";
    else if(number < 0)
        cout << "The number is negative.";
    else
        cout << "The number is zero.";

    return 0;
}
```

Explanation

This program determines whether a number is **positive, negative, or zero**.

- If the number is greater than zero, it is positive.
- If the number is less than zero, it is negative.
- If neither condition is true, the number must be zero.

The **else-if statement** allows checking multiple conditions sequentially.

In this lab, we learned how to use **relational operators, logical operators, and conditional statements** in C++. These concepts are essential for decision-making in programming.

By completing these tasks, we gained practical experience in comparing values, combining conditions, and controlling the flow of a program using **if, if-else, and else-if statements**.